Atopic Dermatitis Monitoring with Wearable Devices

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Health Problem

- Atopic dermatitis (AD), also called eczema, is a common chronic inflammatory skin disease.
- Symptoms include itch, scratching, skin dryness, and barrier dysfunction.
- Around 101.27 million adults and 102.78 million children worldwide have AD.
- Symptoms fluctuate daily, but current assessment relies heavily on self-report.

Motivation

- Wearables enable objective monitoring outside of clinical environment.
- Help track symptom burden and skin condition continuously.

Existing Approaches

- **Wrist-worn actigraphy** can measure night-time scratching and is one of the most established wearable approaches in AD monitoring.
- **Hydration sensing** may reflect skin-barrier condition and dryness, but its clinical use is less established.
- Current challenges include **limited datasets** and **lack of standardized validation methods**.

Sensor type	Application	Measurement accuracy	Limitations	Development stage	Conclusion
Wrist actigraphy	Measurement of itch	High accuracy for night-time scratching	Limited to nocturnal use; does not detect other arm's movements	Widely used in clinical trials	High effectiveness for measuring scratching and sleep disturbances
Smartwatches	Monitoring scratching behaviour	High accuracy (>90% for night-time scratching)	Daytime movements interfere with measurements	Under development for enhanced capabilities	Positive feedback; increased confidence in treatment management
Integrated fabrics	Skin properties (hydration, temperature)	Varies; generally high for specific parameters	Limited by flexibility and sensor integration with textiles	Experimental, some commercial applications	Promising for continuous monitoring of skin health
Microneedle sensors	Long-term biomarker monitoring	High accuracy in stable environments	Potential issues with biofouling; limited long-term data in dermatology	Mainly used in research settings	Effective in theory, needs more data for AD applications

(Khan et al., 2024)

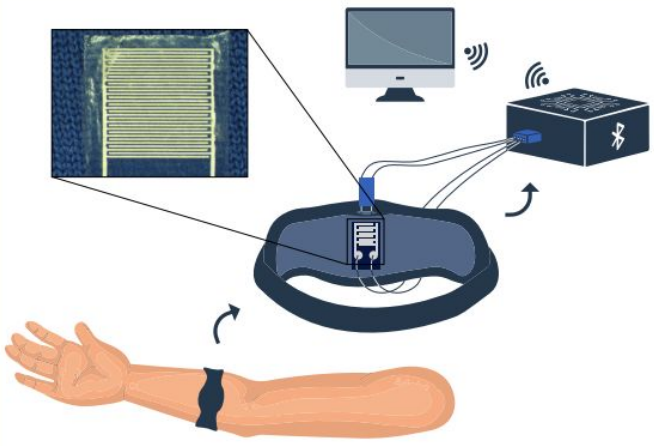
What I want to Accomplish

1. **Replicate paper:** Wearable System Using Printed Interdigitated Capacitive Sensor for Monitoring Atopic Dermatitis in Patients (Alexandar et.al, 2025)
    - Comparing IDC capacitance between lesional and non-lesional skin
    - Validate wearable readings against: 1. Corneometer for skin hydration 2. TEWL for skin barrier loss (optional reference)
2. **Extension model:** extract simple signal features
    - mean, median, standard deviation, range, slope
3. **Exploratory model:** small **1D CNN** on raw 30-second traces
4. Evaluate which method is most effective and most appropriate for this small dataset

Dataset: WearSystemIDCforAD, University of Southampton

- **13 patients**
- **30-second traces sampled at 1 Hz**
- Includes **lesional / non-lesional labels**, **Corneometer**, and **TEWL**

Tools: Pytorch, numpy, scikit-learn



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